THE UNIVERSITY *of* EDINBURGH
School of Mathematics

Financial Mathematics
MATH10003

Saturday 1st May 2021

1300-1500 † *

† **All students**: you have an additional **1 hour** to assemble and submit your PDF.

Final submission deadline: 16:00.

* **Students with a Schedule of Adjustment for additional time in examinations**:
You are entitled to a further fixed additional **1 hour** for this remote examination.

Final submission deadline: 17:00

Attempt all questions

Important instructions

1. Start each question on a new sheet of paper.
2. Number your sheets of paper to help you scan them in order.
3. Only write on one side of each piece of paper.
4. If you have rough work to do, simply include it within your overall answer – put brackets at the start and end of it if you want to highlight that it is rough work.

(1) Let us assume that the stock price today is £80 and it is expected to either increase or decrease by 20% every period over the next three 6-months periods (three period Binomial tree). Also assume that the continuously compounded risk-free rate of interest is 3% per annum and the strike price is £87.

[Total 33 marks]

(a) Verify that the model is arbitrage free.

[1 mark]

(b) Calculate the price of an 18-month European Call option written on non-dividend paying stock S [6 marks] and draw the binomial model [2 marks].

[8 marks]

(c) Calculate the price of an 18-month European Put option written on non-dividend paying stock S.

[6 marks]

(d) Verify if the put-call parity holds using the prices obtained in (b) and (c).

[2 marks]

(e) Calculate the price of an 18-month American Call option written on non-dividend paying stock S [6 marks] and verify if an early exercise is optimal [2 marks].

[8 marks]

(f) Calculate the price of an 18-month American Put option written on non-dividend paying stock S [6 marks] and verify if an early exercise is optimal [2 marks].

[8 marks]

(2) Let $(W_t)_{t\geq 0}$ be a standard Brownian motion and let $a > 0$. Is the the process $(X_t)_{t\geq 0}$ defined as $X_t := aW_{\frac{t}{a^2}}$ a Brownian motion?.

[Total 10 marks]

(3) We have a 4-month European option on a non-dividend paying stock when the stock price is 90, the strike price is 92, the risk-free rate of interest is 4% per annum, and the volatility is 30% per annum. **[Total 35 marks]**

(a) Find the price of the call option. **[5 marks]**

(b) Find the price of the put option. **[5 marks]**

(c) Prove that the Black-Scholes formula satisfy the put-call parity [5 marks] and evaluate it[1 mark].

[6 marks]

(d) What is the mean of the stock price in 2 years?.

[6 marks]

(e) What is the variance of the stock price in 2 years?.

[6 marks]

(f) The investor wants to have a formula to calculate the value at risk, which is given by $v = \inf\{x : P(L \leq x) \geq z/100\}$ where L is a random variable that represent the loss and z is the confidence level. Provide the formula for v. **[7 marks]**

(4) Let $(W_t)_{t\geq 0}$ be a standard Brownian motion and c, d and m positive real constants. Calculate **[Total 10 marks]**

(a) $\mathbb{E}[cW_t + d]$, $\forall t \geq 0$

[2 marks]

(b) $\mathbb{E}[cW_t^2 + d]$, $\forall t \geq 0$

[2 marks]

(c) $\mathbb{E}[uW_{\frac{1}{u}} + sW_{\frac{1}{s}}]$, for any $0 < u < s < \infty$

[2 marks]

(d) $\mathbb{E}[\left(\frac{1}{m}W_{m^2 s} - \frac{1}{m}W_{m^2 u}\right)^2]$, for any $0 < u < s < \infty$

[4 marks]

(5) The following stochastic differential equation,

$$dr_t = \lambda(\mu - r_t)dt + \sigma dW_t,$$

is known in finance as the Vasicek model, where $(W_t)_{t\geq 0}$ is a standard Brownian motion, $\lambda, \mu, \sigma \in (0, +\infty)$ and r_0 is the initial value.

Prove, by applying Itô's formula to $Y_t := e^{\lambda t} r_t$, that

$$r_t = \mu + (r_0 - \mu)e^{-\lambda t} + \sigma \int_0^t e^{\lambda(s-t)} dW_s \quad \text{for every } t \geq 0.$$

[Total 12 marks]

[End of Paper]